SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

MTH312

Examination - January Semester 2026

Real Analysis I

Tuesday, 14 April 2026 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **SEVEN (7)** questions and comprises **FOUR (4)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

(Full marks: 100)

Question 1

For each of the following sequences, either compute the limit if it exists, or otherwise, explain in detail why the limit does not exist (or is infinite).

Question 1a

$$\lim_{n\to\infty}\left(\frac{(-1)^n \times n^2}{(n+5)^3}\right).$$

(5 marks)

Question 1b

$$\lim_{n\to\infty}\left(\sum_{i=1}^{n}\left[i\left(\frac{3}{2}\right)^i\right]\right).$$

(5 marks)

Question 1c

$\liminf\limits_{n\to\infty}\left|n\cdot\sin\left(\frac{n\pi}{6}\right)\right|$, where $|w|$ denotes the absolute value of w.

(5 marks)

Question 2

Define functions

$$f(x)=\begin{cases} x^3+ax^2+bx\ , & x<1 \\ 2ax\ , & x\geq 1\end{cases}$$

and

$$g(x)=\begin{cases} x^4+a\ , & x<2 \\ be^{2-x}-2ax\ , & x\geq 2\end{cases}$$

Determine in exact form the values of a, b that would make $f(x)+g(x)$ continuous on $\mathbb{R}$.

(15 marks)

Question 3

Define a sequence (x_n) as follows: x_1 is a fixed positive real number that is arbitrarily chosen, and

$$x_{n+1} = \frac{(x_n)^2}{2 + x_n}$$

for $n \geq 1$.

Question 3a

Show that (x_n) is a decreasing sequence.

(5 marks)

Question 3b

Use the Monotone Convergence Theorem to show that (x_n) is convergent and find its limit, expressing your answer in terms of x_1 if applicable.

(5 marks)

Question 4a

Compute $\displaystyle\lim_{x \to \sqrt{5}} \frac{25 - x^4}{5 - x^2}$.

(7 marks)

Question 4b

Compute $\displaystyle\lim_{x \to 0^+} \left(1 + \frac{x}{2}\right)^{3/x}$.

(6 marks)

Question 4c

Compute $\displaystyle\lim_{x \to 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$.

(7 marks)

Question 5

If A and B are subsets of $\mathbb{R}$, define

$$d(A,B) = \inf\{|x-y| \ : \ x \in A, y \in B\}.$$

Question 5a

Suppose that $(X_n)_{n=1}^{\infty}$ is a sequence of sets and X is a set, and all of them are subsets of $\mathbb{R}$. Show that

$$\lim_{n\to\infty} d(X_n, X) = 0$$

if and only if there exists a sequence (a_n) such that $a_n \in X_n$ for every positive integer n, and a sequence (b_n) such that $b_n \in X$ for all n such that

$$\lim_{n\to\infty} (a_n - b_n) = 0.$$

(12 marks)

Question 5b

Give an example of a sequence (A_n) of open intervals and an open interval A such that A_i is disjoint from A_j for $i \neq j$, and A_i is disjoint from A for each i, and

$$\lim_{n\to\infty} d(A_n, A) = 0.$$

(3 marks)

Question 6

Apply an $\epsilon - \delta$ proof to demonstrate that the function $f(x) = \dfrac{1}{1+\sqrt{x}}$ is continuous at the point $x = 2$

(15 marks)

Question 7

Suppose that $f : \mathbb{R} \longrightarrow \mathbb{R}$ is a continuous function such that for every y in the range of f, there are at most two values of x for which $y = f(x)$. Show that there is some $z \in \mathbb{R}$ such that $z = f(x)$ for one and only one value of x.

(10 marks)

----- END OF PAPER -----